MEDNOVA HEALTHCARE PROJECT

Prompt Engineering & BRD Development Methodology

An executive framework for role-prompted requirements engineering, Responsible AI governance, enterprise architecture, healthcare analytics, and digital experience design.

DOCUMENT CLASS	Executive Methodology
PRIMARY AUDIENCE	CEO • CTO • CIO • CISO • Chief Data/AI Officer • Enterprise Architecture
DELIVERY CONTEXT	AI-Powered Healthcare Medical Institution BRD

Purpose. Establish a disciplined, auditable prompt-engineering methodology that converts executive healthcare objectives into traceable BRD requirements, governance controls, architecture, workflows, data structures, analytics, and experience specifications.

Executive Summary

MedNova's BRD methodology uses advanced role prompting to transform a complex healthcare technology mandate into four controlled, sequential workstreams. Each stage assigns the AI a precise enterprise role, defines explicit boundaries, enumerates required outputs, and establishes quality and governance expectations. The approach is designed to improve completeness, consistency, traceability, and executive readability while preventing uncontrolled clinical decision-making or unsupported claims of regulatory compliance.

Four-Stage Prompt Architecture

Stage	Role-Prompt Domain	Primary Output	Executive Value
01	Business & Functional Requirements	BRD foundation, scope, stakeholders, functional/non-functional requirements	Aligns investment intent with measurable requirements and success criteria.
02	Responsible AI, HIPAA, Security & LLM	Privacy, RBAC, audit, governance, AI safety, validation, customized LLM controls	Makes trust, accountability, human oversight, and data protection explicit.
03	Enterprise Architecture & Workflows	Layered architecture, data flows, RBAC, patient/hospital/admin/AI workflows	Connects business requirements to an implementable enterprise operating model.
04	Data, Analytics, Dashboards & UX	Patient model, 20 synthetic records, analytics, reporting, design system, traceability	Translates requirements into measurable information products and a governed user experience.

Methodology

The methodology follows a progressive decomposition model: business intent is established first; risk, privacy, security, and Responsible AI controls are then overlaid; enterprise architecture and end-to-end workflows operationalize those requirements; and the final stage defines data, analytics, reporting, dashboards, and digital experience specifications. This sequencing reduces ambiguity because each prompt inherits the controlled context established by the prior stage.

Core Prompt-Engineering Techniques

Multi-role expert framing

Assigns complementary enterprise personas - business architecture, clinical informatics, security, AI governance, solution architecture, data architecture, analytics, and UX - to increase domain coverage without losing accountability.

Constraint prompting

Explicitly prohibits website construction and application code, keeping the AI focused on requirements and architecture documentation.

Structured decomposition

Breaks a broad healthcare transformation problem into bounded sections with named deliverables, matrices, workflows, and requirement identifiers.

Output-schema prompting

Specifies tables, matrices, IDs, acceptance criteria, architecture layers, Mermaid diagrams, and traceability structures so outputs are consistent and reviewable.

Risk-aware prompting

Separates AI assistance from clinical decision-making and requires validation, human review, approval, monitoring, and auditability.

Synthetic-data prompting

Requires demonstration records and medical imagery to be synthetic/educational, reducing the risk of implying real patient evidence or exposing real PII/PHI.

Traceability prompting

Maps assignment/business requirements through BRD sections, architecture, workflow, data, controls, UI, and analytics to create an auditable line of sight.

Stage 1 | Business & Functional Requirements

Role construct: Enterprise Healthcare Business Requirements Architect

Establish the business case and requirements baseline for patient management, healthcare operations, administration, reporting, analytics, AI-enabled education, and digital experience.

Prompt Design Pattern

Role → Boundary → Required domains → Structured outputs → Governance controls → Quality standard

Expected Deliverables

- Project vision and measurable objectives
- In-scope, out-of-scope, and future scope
- Stakeholder matrix and access expectations
- Functional requirements with professional IDs and acceptance criteria
- Measurable non-functional requirements
- Page-level functional scope and success criteria

Control intent. This stage creates the authoritative requirements baseline. It uses requirement IDs such as FR-PAT, FR-ADM, FR-ANA, and FR-AI to support downstream traceability.

Stage 2 | Responsible AI, HIPAA, Security & Customized LLM

Role construct: Healthcare Responsible AI, Security & Governance Architect

Overlay the BRD with privacy, security, governance, AI safety, human oversight, and controlled LLM requirements.

Prompt Design Pattern

Role → Boundary → Required domains → Structured outputs → Governance controls → Quality standard

Expected Deliverables

- HIPAA-aligned privacy/security requirements without claiming certification
- PII/PHI classification, minimization, masking, encryption, redaction, and de-identification
- RBAC based on least privilege, need-to-know, and separation of duties
- Comprehensive audit logging and data governance
- Configurable retention requirements subject to applicable policy and law
- AI safety, validation, approval, incident management, monitoring, and accountability
- Controlled text-to-image, text-to-audio, narration, and customized LLM/RAG requirements

Control intent. The prompt explicitly prevents AI from independently making high-risk clinical decisions and requires human/clinical approval for high-risk medical content.

Stage 3 | Enterprise Architecture & End-to-End Workflows

Role construct: Principal Healthcare Enterprise Solution Architect

Convert business and governance requirements into a coherent conceptual architecture and controlled operating workflows.

Prompt Design Pattern

Role → Boundary → Required domains → Structured outputs → Governance controls → Quality standard

Expected Deliverables

- Presentation, application, AI, data, and security layers
- Conceptual enterprise architecture with trust boundaries
- Patient and hospital lifecycle workflows
- Administrative and RBAC workflows
- Responsible AI workflow with PII/PHI protection and approval gates
- Data flow and Replit integration architecture
- Customized LLM/RAG architecture with security and governance boundaries
- Architecture decision matrix

Control intent. The architecture applies Security by Design, Privacy by Design, Responsible AI by Design, Zero Trust, least privilege, defense in depth, human-in-the-loop, auditability, modularity, resilience, and observability.

Stage 4 | Healthcare Data, Analytics, Dashboards & UX

Role construct: Healthcare Data Architect, Analytics Architect & Digital UX Design Director

Complete the information model and define how trusted healthcare information is measured, reported, visualized, and experienced.

Prompt Design Pattern

Role → Boundary → Required domains → Structured outputs → Governance controls → Quality standard

Expected Deliverables

- Complete patient data model with PII/PHI/clinical/operational/audit/analytics classification
- Exactly 20 synthetic demonstration patient records
- Severity, health, diagnosis, admission, discharge, check-in, medication, ER, monitoring, and operational analytics
- Executive dashboard KPIs and visualization requirements
- AI-generated reporting controls and approval metadata
- Modern healthcare design system, responsive UX, page specifications, and accessibility
- Final end-to-end requirements traceability matrix

Control intent. The final stage ensures that the BRD can be evaluated not only as a requirements document but as a traceable blueprint for governed analytics and user experience.

Governance & Safety Considerations

Control Area	Prompt Engineering Mechanism	BRD Effect
Regulatory posture	Uses HIPAA-aligned requirements and explicitly rejects unsupported certification claims.	Creates a defensible compliance requirements posture for later formal validation.
PII/PHI protection	Requires classification, minimum necessary access, encryption, masking, de-identification, RBAC, and audit.	Embeds privacy and security into architecture and workflow design.
Clinical AI safety	Separates AI assistance from clinical decision-making; requires review and approval.	Preserves human accountability for high-risk medical content.
Synthetic demonstrations	Labels patient records and generated scan-style imagery as synthetic/educational.	Prevents demonstration content from being represented as real patient evidence.
AI accountability	Requires validation status, approval status, reviewer, version, timestamp, audit record, monitoring, and incident management.	Creates governance evidence across the AI content lifecycle.

BRD Traceability Model

Traceability is treated as a design requirement rather than an after-the-fact reporting exercise. The final methodology requires each source requirement to be mapped across the BRD and into the enterprise design.

Business Requirement → BRD Requirement → Architecture → Workflow → Data → Responsible AI Control → UI/Page → Analytics

Executive Quality Standard

The four-stage prompt system is intentionally engineered to produce documentation that is enterprise-grade, healthcare-focused, internally consistent, security- and privacy-conscious, Responsible-AI aligned, and suitable for executive review. Its principal strength is controlled progression: each prompt narrows ambiguity, formalizes outputs, and adds traceable controls before the next architectural layer is introduced.

Executive Conclusion

MedNova's prompt-engineering approach demonstrates how generative AI can be governed as a structured requirements-engineering capability rather than used as an unconstrained content generator. Advanced role prompting establishes domain expertise; explicit boundaries constrain scope; output schemas make deliverables reviewable; and safety, privacy, audit, approval, and traceability requirements establish enterprise discipline. The resulting methodology provides leadership with a clear line of sight from strategic healthcare objectives to BRD requirements, Responsible AI controls, enterprise architecture, operational workflows, information models, analytics, and digital experience.